Data Mining and Spatial Analysis of NASA Meteorite Landings
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Introduction

Meteorites are pieces of material from space that survive their passage through Earth's atmosphere and land on the surface of the planet. They are important because they can carry information about asteroids, planets, and other objects in the solar system. By studying meteorites, scientists can learn more about what these objects are made of, where they may come from, and how materials from space interact with Earth.

Analyzing meteorite landing data is interesting because it allows us to view meteorites not only as individual scientific objects but also as part of a larger global dataset. NASA's Meteorite Landings dataset includes records of meteorites found around the world, with information such as name, classification, mass, year, fall status, latitude, and longitude. Because the dataset includes location information, it can be used to study spatial patterns and compare how meteorite records are distributed across different regions of the world.

This type of analysis can lead to real-world discoveries because patterns in the data may reveal information that is not obvious from looking at raw records alone. For example, meteorite data can show whether certain areas have more recorded meteorites, whether high-mass meteorites appear in specific regions, or whether meteorite discoveries change over time. These patterns can help researchers ask better questions about both natural meteorite distribution and human data collection practices.

Spatial patterns are especially important in this project because meteorite records are connected to geographic coordinates. By using latitude, longitude, continent, distance from the equator, and time period, we can study whether meteorites appear evenly across the world or whether some regions are more strongly represented than others. However, the dataset may also contain bias. Meteorites may be more likely to be found and recorded in areas with more people, easier terrain, more scientific activity, or better reporting systems. Because of this, a high number of meteorite records in one region does not always mean more meteorites actually fall there. It may also reflect how humans collect and report the data.

This project focuses on three main research questions:

- **Research Question 1:** Is there a statistical relationship between meteorite mass and distance from the equator?
- **Research Question 2:** In the clustering results, which continents are overrepresented in each cluster, and does that distribution change between the historic and modern periods?
- **Research Question 3:** Are there distinct high-mass meteorite clusters with specific geographic signatures?

To answer these questions, we cleaned and preprocessed the NASA Meteorite Landings dataset, prepared important features such as mass and geographic location, and applied clustering methods to search for meaningful groups in the data. We also evaluated the clustering results and used visualizations to better understand the patterns produced by the analysis.

Data Mining Tasks

Dataset Description: The dataset used in this project is NASA's Meteorite Landings dataset, which is provided as a CSV file. The dataset contains around 45,000 meteorite records, with each row representing one meteorite. Each record includes information about the meteorite's identity, classification, mass, fall status, year, and geographic location. Since our project focuses on spatial analysis and clustering, the most important parts of the dataset are the meteorite's mass, latitude, longitude, year, and whether it was classified as "Fell" or "Found."

Dataset Features: The original dataset includes several columns, but the main ones we worked with are:

- **Name and ID:** These identify each meteorite record. They are useful for tracking and storing the data, but they were not used directly for clustering because they do not provide numeric or spatial meaning.
- **Recclass:** This column describes the meteorite's classification or mineral type. We kept this column in the dataset description because it is part of the original NASA data, but we later decided not to use it directly in clustering because it is a non-metric categorical value with many different categories. Because of that, it was difficult to use in a meaningful way for our geographic clustering goals.
- **Mass (g):** This gives the mass of each meteorite in grams. This feature was important for our project because one of our research questions looks at whether mass has a relationship with distance from the equator. Since meteorite masses can be extremely skewed, this column needed extra preprocessing before analysis.
- **Fall:** This tells whether the meteorite was observed falling, labeled as Fell, or was discovered later, labeled as Found. This was important because we wanted to compare patterns between meteorites that were directly observed and meteorites that were found later.
- **Year:** This gives the year connected to the meteorite record. We used this to help separate meteorites into historic and modern periods so we could compare whether clustering patterns change over time.
- **Geolocation:** This includes latitude and longitude values. These were some of the most important features in the project because they allowed us to study where meteorites were recorded around the world and to create spatial cluster visualizations.
- **Distance from the equator:** This was created as a new feature so we could study whether meteorite mass has a relationship with how far the meteorite was from the equator.
- **Continent or region information:** This was created from the geographic location so we could compare meteorite patterns across different parts of the world.

Inputs: The main input for this project is the NASA meteorite CSV file. We used the raw meteorite landing records and then cleaned and processed them into a more usable dataset. The main input features we focused on were:

- Meteorite mass
- Log-transformed mass
- Latitude and longitude

- Fall status, meaning Fell or Found
- Year
- Distance from the equator
- Continent or region information created from the geographic location
- Historic and modern time period labels

Outputs: The outputs of this project include cleaned datasets, processed datasets, clustering results, visualizations, and summary tables. The main outputs are:

- A cleaned meteorite dataset saved as a CSV file
- A processed dataset with new features prepared for clustering
- Cluster assignments for different subsets of the data
- Clustering plots saved as PNG images
- Tables showing cluster summaries and continent representation
- Visual comparisons between Found, Fell, Historic, and Modern meteorite clusters

Excluded Features: Even though columns like name, ID, and recclass are useful for describing the original dataset, we decided not to use them directly in the clustering process. Name and ID are identifiers, so they do not help the model find meaningful numeric or spatial patterns. Recclass could be useful in a different type of analysis, but for this project, it was difficult to cluster with because it is a non-metric categorical value with many categories. Because of that, we focused more on mass, location, fall status, year, and geographic features.

Main Data Challenges: The dataset had several problems that needed to be handled before analysis. Some records had missing values, incomplete location information, and extreme mass values. These issues mattered because they could affect the clustering results and make the patterns harder to understand.

One specific issue was **Meridiani Planum**, a meteorite record with a longitude value outside the normal Earth range of -180 to 180. After checking it, we found that this record was from **Mars**, not Earth, so it had to be handled carefully for our Earth-based spatial analysis.

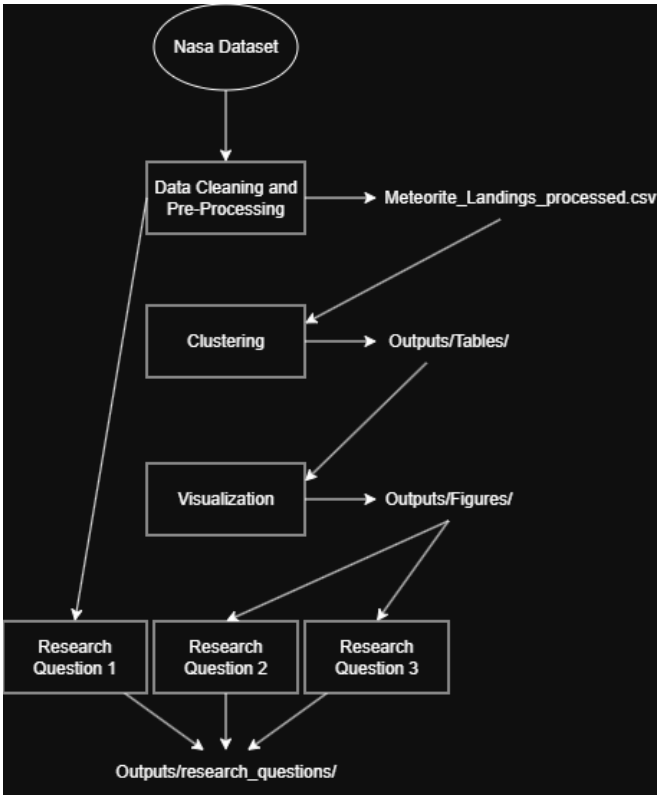
Another major issue was the **mass column**. Meteorite mass was extremely skewed, with a skew value of around **76.91**. After applying a log transformation, the skew dropped to about **0.91**, which made the data easier to use for clustering and analysis.

The dataset was also very imbalanced between **Fell** and **Found** meteorites. There were about **44,609 Found records** and only **1,107 Fell records**. This matters because Found meteorites may reflect human collection patterns more than natural meteorite fall patterns.

Finally, the data was not evenly distributed across continents. Antarctica had the largest number of records, with about **22,099**, while other continents had much smaller counts. This suggests possible collection bias, since some regions are easier to search or preserve meteorites better.

Technical Approach

Summary: Our approach to this task was split into 2 main sections that we worked on iteratively, then a final 3rd section to wrap everything up. We first experimented with code in jupyter notebooks, using the fact that we could run individual cells to test specific aspects of the pipeline, as well as ease of use and sharing between teammates. Then, once we "completed" a section in the notebook, we would turn to the Python script that we use as the actual finished product for outputting results. Finally, we added specific code for answering our more particular research questions, rather than a general data overview that we had before.



Data Flow: We start with the Nasa meteorite csv dataset, and run it through a series of data cleaning functions to get rid of empty/invalid values and duplicate entries.

We then run through pre-processing, transforming heavily skewed data like mass (g), and turning it into mass log. We added additional features we thought we might need, such as distance from the equator, a binary form of the Fall feature for clustering later on, and more. Then, in clustering, we first subset the existing data into 7 different sections.

- The full, primary dataset
- Fell only
- Found only
- Valid geo data
- Land only
- Historic (pre 1950)
- Modern (post 1950)

Once we had our subsets, we ran them through our K-means algorithm, using silhouette scores to evaluate what the optimal k was for each subset. This was done via a function, run_clustering_and_eval, which did the following:

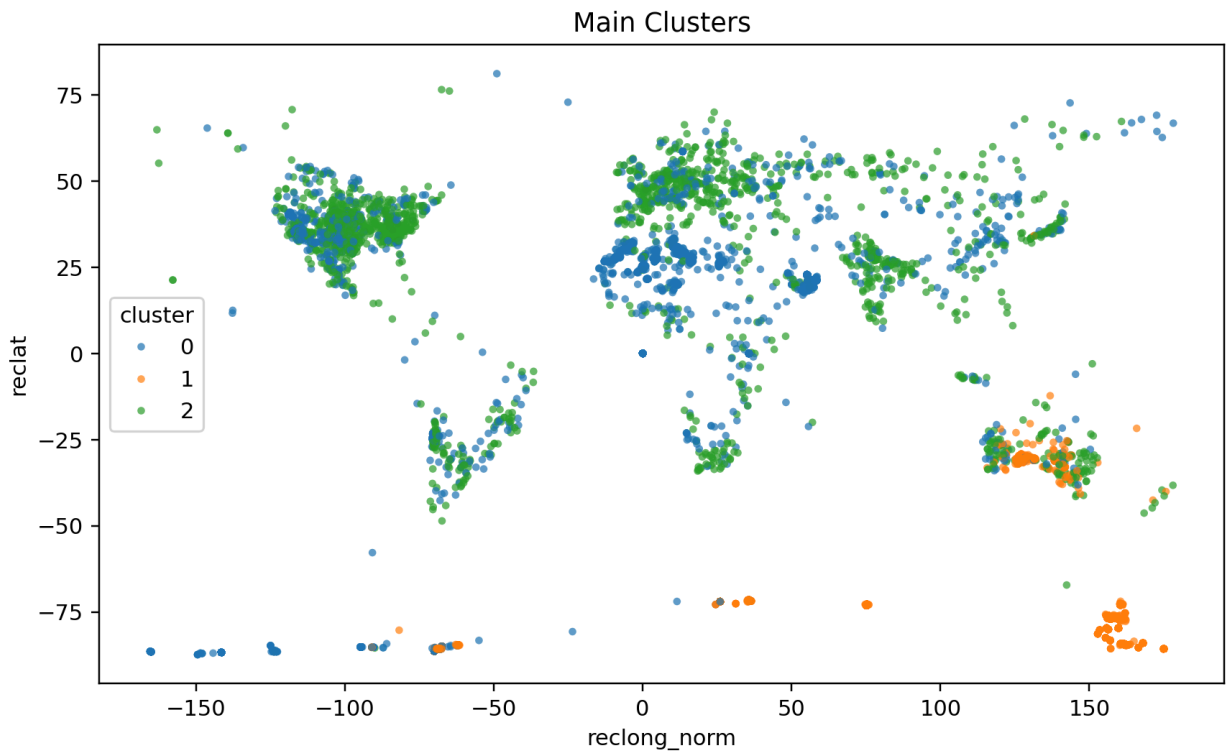
- Using the given subset and feature columns, first filter for usable rows (numeric).
- Standardizes features using sklearn's StandardScaler function.
- Loops over each K in the range, calling sklearn's KMeans function on each.
- At each k, scores with a silhouette score (using a subset scorer, as otherwise it would take hours to complete fully for all of our clustering)
- Selects the best k
- Takes the original rows that were clustered and adds a cluster ID to each
- Returns the best k, best K model, scores table, scalar, clustered data, and the feature set.

Using this function on each of our 7 subsets from before, we can get clustering data for all of the subsets we might need in the future for our research questions. Interestingly, the vast majority went unused after visualization was complete, but I believe it is better to plan for needing more than you

actually do. Once we completed clustering, we needed to build our cluster center data into formatted tables that display each cluster's average values. This helps immensely later on for our research questions!

After the clustering section, it's time for visualization. Using another static function, we went through each saved cluster.csv and output two types of clustering comparisons as a general baseline. The function accepted x and y columns, which acted as the type of cluster we wished to make / analyze.

Example: Our main dataset is clustered on latitude and longitudinal values, clearly outlining continents!



Research Question Analysis: Once we completed our general visualization and initial mining of data, we focused on the specific answers we need for our research questions. At this point, our pipeline contained our 4 sections, data cleaning, preprocessing, clustering, and visualization on top of a main section that was used to call the various functions and pipeline calls for individual sections.

We then determined the best way to get the information we needed for our research questions would be to make a 5th section after visualization, where we used all the outputs from the pipeline to answer any questions we might have, with additional figure code and math if needed. As a reminder, our 3 questions to answer were:

1. How does the mass of a meteor correlate with its distance from the equator? (Statistics)
2. Which continents are overrepresented in each cluster, and does that distribution change between historic and modern periods? (Clustering)
3. Are there distinct high mass clusters with a specific geographic signature? (Clustering)

Answering Research Question 1: For question 1, we did a rather straightforward correlation analysis on the processed dataset from step 2 in our pipeline. Using the outputted .csv from pre-processing, we sliced out our log_mass and dist_equator_km features, and filtered for valid geolocation rows, excluded placeholders, and got rid of missing values. We then ran two correlation calculations between log_mass and dist_equator_km:

- Pearson correlation to see a linear relationship
- Spearman correlation to see a rank/monotonic relationship.

Finally, we generated a hex-bin figure to visualize any trends we might have seen through the data.

Answering Research Question 2: For our second research question, we primarily needed 2 things. Our modern_baseline_clustered.csv and historic_baseline_clustered.csv, which were just the modern and historic subsets clustered with the baseline featuresets. Once we had these two csv's, we just needed the cluster and continent features. Once we loaded the clustered datasets, we once again removed any invalid rows, then for each period (modern/historic) we computed the overall continent distribution in that period, as well as the continent distribution inside each cluster.

Then, using this info, we found the overrepresentation ratio by dividing (continent share in cluster) by (continent share overall). If the ratio is greater than 1, then it was an overrepresented continent. We then visualized the modern vs historic groupings in side by side heatmaps of the ratios, so we could see our results.

Answering Research Question 3: Finally, for the third question, we needed to use our main_baseline_clustered.csv and main_baseline_centers.csv. After loading the csv's, we defined a high-mass cluster using centroid log_mass to compute a percentile threshold (90%) to then get the clusters whose centroid was above said threshold. We then build a per-cluster signature table from the row data, and filtered the table to show only the high-mass clusters, then created a new cluster figure using the geo-location coordinates so we got a map of the meteorite locations around the globe, to then see what the dominant continent was, if any.

Evaluation Methodology

Dataset: We used the NASA meteorite landings dataset CSV, from the official nasa.gov site: Meteorite Landings - Dataset - NASA Open Data Portal.

- The dataset had 45,716 rows
- Key features we used from the dataset were mass, year, geo location, our derived distance from equator in kilometers, our derived continent and country features, and found vs fell.

Data Challenges:

- Missing and noisy values in mass, year, and location.
- Mass was extremely skewed, solved with a log1p transform in preprocessing.
- We had a few invalid/placeholder coordinates (0,0) and out-of-range longitudes
- There was some spatial ambiguity for ocean sections, resulting in them all being categorized as simply ocean.
- Potential observational bias due to observers of meteorites being in generally populated areas.

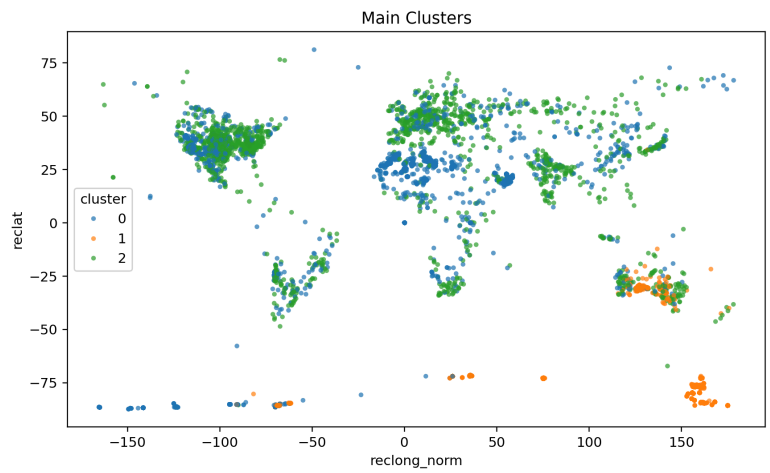
Evaluation Metrics:

- For clustering quality:
    - Silhouette scores to select our KMeans models and k values, as it measures the separation vs compactness of clusters, which is important since better separated clusters make it much more visually distinct during analysis.
    - Inertia for compactness in each cluster, since tighter groups give us a more coherent individual cluster for each.
- For research question 1:
    - Pearson correlation for the linear association strength
    - Spearman correlation for rank/monotonic association, which is a bit more solid then Pearson on its own.
    - P-values for both, in this case, were so close to 0 it was not in the integer limit for python, and was displayed as 0.0 due to underflow. This means there is a high likelihood our correlation results were NOT just noise, and there is indeed some reasoning behind it.
- For research question 2:
    - Overrepresentation ration found by dividing the continent share in clusters by the continent share in the period, which gives us clear examples of the expected data vs its overrepresentation, if any.
    - Modern vs historic comparison done via side by side heatmap so we could properly evaluate which had overrepresented continents and their differences.
- For research question 3:
    - Centroid-based threshold that we decided/tested with, landing at 90% due to heavy skew despite earlier transformation,
    - Cluster signature statistics like mean_log_mass to get averages for cluster average mass splits.

Results and Discussion

Our data began as a simple csv, with holes and many invalid values. One such value was the geo location for the meteorite Meridiani Planum, with a longitudinal value of 354. Interestingly, every other meteorite in the dataset that did not just have missing values for geo location had a valid number set, except this one. When we researched it, it turns out Meridiani Planum was actually discovered on mars, found in the dirt. In the data, it is also listed under the found value, meaning it was indeed dug up. As to why it had an odd longitude range, or why it appeared in this dataset at all despite being seen on Mars, we aren't sure.

The idea is that our data was flawed, albeit a lot less than I expected it to be, likely due to the fact that it was from an official government site and provided by NASA. Once we realized it had these small errors, we dove into more detailed data processing, even using an external dataset from natural earth, to map our meteorite locations to actual landmasses on Earth, and extract a continent AND country feature into our processed dataset. This was actually used in solving two of our research questions, so it was very helpful to do so.



Before we started working on answering the specific research questions, we wanted to observe our clustering data and produce as much as we could ahead of time for use in the questions later on.

Main Clusters Centered Table:

Log_mass	reclat	reclong_norm	year	cluster
4.69	6.16	1.6	2000	0
2.84	-75.99	108.16	1990	1
8.7	27.84	-19.30	1904	2

Aside from our research questions, this was our most interesting clustered plot, showing the full dataset of meteorites, split into k=3 clusters after silhouette scores determined that was the optimal k value to use. Interestingly, we have an average log_mass of one of our 3 clusters at double the size of the next largest, despite the mass transformations we did earlier to try and lower this skew towards heavier meteors. It's also worth noting (as this was used in answering question 3), that this extremely high mass cluster was also focused on North America, in our latitude and longitudinal clustering map.

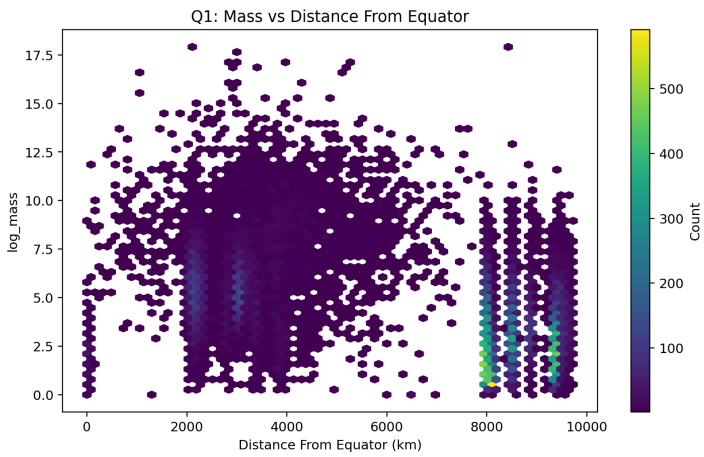
Research Questions: Finally, we now answer our research questions using the numerical data as well as generated plots, we can analyze our questions and get solid, satisfactory answers to each.

Question 1: Is there a statistical relationship between meteorite mass and distance from the equator?

- Yes, there is. By observing pearson and spearman correlations on the transformed mass and distance from equator features, we could see a clear, if slight, trend towards meteor mass being negatively correlated with distance from the equator. Meaning that the closer to the equator meteors were found/seen, the heavier they tend to be.

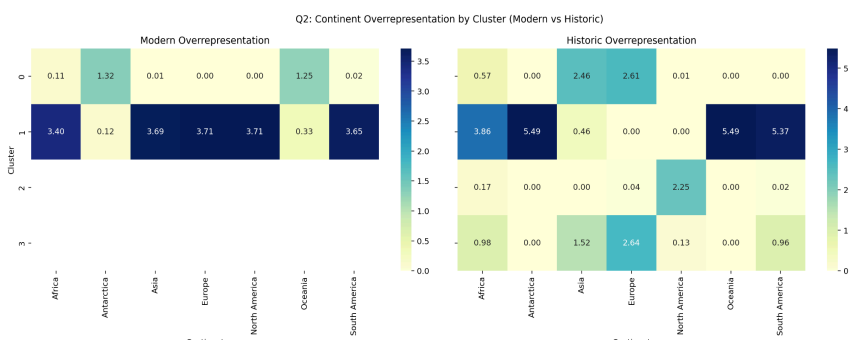
n_rows	pearson_r	pearson_p	spearman_rho	spearman_p
32068	-0.513995	0.0	-0.388739	0.0

- It is worth noting that both of the p-values for pearson and spearman were 0.0, due to what we believe to be underflow. Assuming this is the case, this means we had a clear lack of noise pollution in our results, meaning it is likely that, however small, this correlation DOES exist and is not the cause of randomness in our dataset.



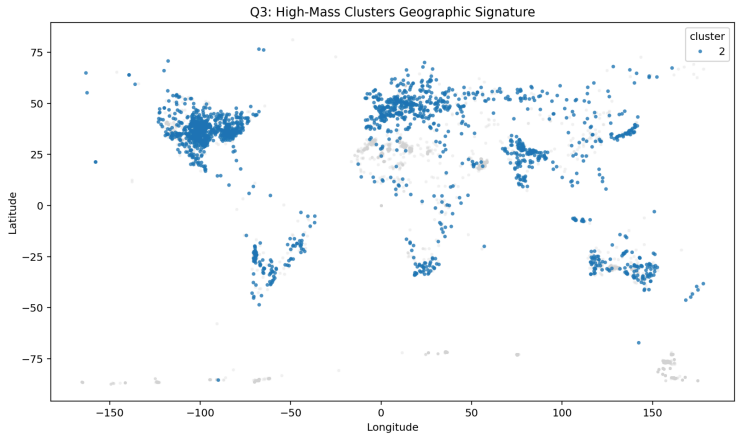
Question 2: In the clustering results, which continents are overrepresented in each cluster, and does that distribution change between the historic and modern periods?

- We can clearly see that in modern times, it is tied between Europe and North America for clustering overrepresentation, while historically, the most overrepresented continent was Oceania. This tells us that there are a lot more meteorites being found in Europe and North America, while in the past, primarily they were found in Oceania, compared to the rest of the world.



Question 3: Are there distinct high-mass meteorite clusters with specific geographic signatures?

- Regarding question 3, we observed "high-mass" clusters (which we determined to be those in the 90th percentile) and clustered them to determine our answers. Once again we see a similar result to our modern clustering overrepresentation numbers, with North American and Europe taking the crown for this section, with North America at 856 rows, and Europe with 408.



Conclusions: Overall, our data mining project was a success, in our eyes. However, this does not mean we had no problems at all. Regarding what went well, first:

- Generating clusters through a static function that we could reuse on different subsets let us change as little as possible between clusters. Using this, we were able to get a lot of clustering data, then also plot out the various features we wanted to compare.
- Our dataset was very clean from the get-go, and we all but ignored all the text-based features it came with, Initially, this meant our data was extremely easy to process and clean, only requiring a log transformation to normalize our mass values (they were still very skewed, but significantly less than they were initially)
- Our Jupyter notebook setup allowed us to iterate on code design and make changes without having to rerun the entire pipeline start to finish, which helped massively when doing clustering.

Now, for what went poorly:

- First of all, our clustering was flawed. We used silhouette scores to determine optimal k values to use for our KMeans call, but we had a few problems with using silhouette. First, we should have used seeded KMeans runs, as those would have allowed us to keep more consistent outputs as we iterated through clustering decisions and reruns. We did not realize how important seeding runs were, and having random starter centroids every time messed us up a little bit.
- Our clustering was also less accurate than it could have been. Initially, we ran uncapped silhouette scoring, running through the whole dataset on every possible K, but given the sizes we reached, this took an extremely long time. On our first test run, we left it running for 60 minutes and nothing happened, so we determined we needed to lower the amount of scoring we allowed, to keep our runs down to a more reasonable time.
- We believe it would have also been better to split our main script into multiple files like we originally planned. We had multiple jupyter notebooks, but we moved them all into one script, which resulted in more bloat and hard to read code then we thought initially. Not too big a deal, but not ideal either.

All in all, in future similar projects, we would first fix our clustering methodology. It was crude, and could have been significantly better. We should have done additional clustering, and maybe compared it as well, perhaps hierarchical, then seeing if it resulted in significantly different plots to our regular KMeans algorithm.

References:

Natural Earth. *Admin 0 – Countries (1:10m Cultural Vectors)*. Natural Earth, Natural Earth » Blog Archive » Admin 0 – Countries - Free vector and raster map data at 1:10m, 1:50m, and 1:110m scales

National Aeronautics and Space Administration (NASA). *Meteorite Landings*. NASA Open Data Portal, Meteorite Landings - Dataset - NASA Open Data Portal (Filtered: Utm source)